



Microsoft Windows Surface

device won great acclaim at an intercollegiate contest. It is an infrared-based gesture interface device that can be used as a remote control for televisions, air-conditioners, computers or any other appliance fitted with the Air Glove infrared interface. The device has an advantage over radio frequency based devices and can operate multiple appliances simultaneously. Moreover, the infrared technology has pinpoint accuracy, making it more viable.

While the team demonstrated the concept with a large form-factor device, they are now developing a wrist-watch model.

Betting big on touch and gesture user interfaces

Microsoft Surface has already become a famed technology. The 360-degree interface uses PixelSense technology to see and respond to touch and gestures of real-world objects. It accepts over 50 simultaneous inputs, and has been used in the 102cm (40-inch) Samsung SUR40, which can be embedded on tables and walls.

Kinect, another Microsoft technology, has also made a mark. This motion-sensing input device for Xbox 360 and Windows PCs uses a webcam-like peripheral and associated software that lets users control and interact with the device through gestures and spoken commands. Microsoft hopes that Kinect will play a huge role in industries like healthcare, apart from interactive

television and computer systems.

During a talk at CES, Steve Ballmer commented that Windows 8's Metro user interface would drive a new magic that will make "one plus one equal three." The Metro features a touch-enabled, tiled Start screen. Each tile represents an application and display information

related to that application, such as the number of unread messages for an email application or the document currently being used on a wordprocessor. Metro applications can share information between them, and Microsoft hopes it to revolutionise cross-device user interfacing.

Indeed, there is a lot of focus on gesture-based computing these days, and hardware and software products are coming up to enable gesture-based user interfaces. San Francisco based Leap has introduced a small plastic device for enabling gesture applications on Mac. You need to simply plug the Leap into a USB port, load the Leap Motion software and calibrate the device by waving your arms as instructed. Having done that, the device becomes capable of tracking the motion of your hands or fingers very precisely (in the scale of around 1/100th of a millimetre), within a space of around 0.22 cubic metres (8 cubic feet).

Leap hopes that more developers will create software to take advantage of this gesture capability. It has provided sample applications and a software development kit on its website.

Pranav Mistry's SixthSense is another interesting gesture computing technology. His work with MIT's Media Lab involves converting natural hand movements into digital information, to interact with computer systems. The SixthSense prototype com-

prises a pocket projector, a mirror and a camera. The hardware components are coupled in a pendant-like device, and connected to the mobile computing device in the user's pocket.

The projector projects visual information enabling surfaces, walls and physical objects around us to be used as interfaces, while the camera recognises and tracks the user's hand gestures and physical objects using computer-vision based techniques. The software program processes the video stream data captured by the camera and tracks the locations and movements of the coloured markers at the tip of the user's fingers. This information is interpreted into gestures, which, in turn, act as instructions to an application.

SixthSense supports multi-touch and multi-user interaction. For more information about it, read the interview with Pranav Mistry published in EFY's February 2012 issue (<http://bit.ly/yTj8Vg>).

Sparsh, another of Mistry's projects, is equally interesting. In an interview with EFY, Mistry explains "Sparsh is an interaction method that lets users conceptually transfer media from one digital device to their body, and pass it to another digital device through simple touch gestures. In Sparsh, we are playing with the perception of the user. When a user touches any data, it appears as if it becomes part of the user. But actually the user acts just as a token or ID and the data gets copied in a particular folder that belongs to the user's ID in the cloud, and stays there.

The data is matched with the token of the user in the cloud. Now when the user touches another media or device where he wishes to paste the data, the device accesses the data corresponding to the user's ID in the cloud.

"It is something like e-mail in concept, which can be checked from anywhere. But if I show something as simple as an e-mail to my grandma, she may wonder how I am able to check the same e-mail in office that I checked at home. She may ask if I